

Multiple Choice (10 marks)

1. A water pipe in which water pressure is 80 lb/in.^2 springs a leak, the size of which is $.01 \text{ sq in.}$ in area. How much force is required of a patch to "stop" the leak?
 - (a) 8 oz
 - (b) 0.008 lb
 - (c) 8 kg
 - (d) 800 lb
 - (e) .8 lb
2. A nucleus of mass M emits a γ ray of energy $E_{\gamma \text{ref}}$. The nucleus was initially at rest. What is
 - (a) $E_{\gamma}^2 / 4Mc^2$
 - (b) $2E_{\gamma}^2 / Mc^2$
 - (c) E_{γ} / Mc^2
 - (d) $3E_{\gamma}^2 / 2Mc^2$
 - (e) $E_{\gamma} / 4Mc^2$
3. When 100 J are put into a device that puts out 40 J of useful work, the efficiency of the device is
 - (a) 25%
 - (b) 100%
 - (c) 60%
 - (d) 50%
 - (e) 75%
4. Find the minimum energy of a simple harmonic oscillator using the uncertainty principle.
 - (a) $\frac{h\omega}{4\pi}$
 - (b) $m\omega^2/2$
 - (c) $\frac{h}{m\omega^2}$
 - (d) $\frac{h^2}{2\pi m\omega}$
 - (e) $h^2/m\omega$
5. What is the total charge in coulombs of 75.0 kg of electrons?
 - (a) $-2.05 \times 10^{13} \text{ C}$

- (b) $-1.32 \times 10^{13}\text{C}$
- (c) $-1.98 \times 10^{13}\text{C}$
- (d) $-0.98 \times 10^{13}\text{C}$
- (e) $-1.44 \times 10^{13}\text{C}$

True / False (10 marks)

Answer True or False

6. True or False: An electron can reach a speed of $1.8 \times 10^7 \text{ m/s}$ by being accelerated across a potential difference of 100 V.
7. True or False: A uniformly charged conducting sphere with a diameter of 1.2 m and surface charge density of $2 \times 10^{-6} \text{ C/m}^2$ has a total charge of $2.5 \times 10^{-5} \text{ C}$.
8. True or False: An atom with filled $n = 1$ and $n = 2$ energy levels contains a total of 12 electrons.
9. True or False: The focal length of the lens is -20 inches, indicating that it is a diverging lens.
10. True or False: The specific heat capacity of sand is always equal to that of vegetation, leading to the same rate of temperature change.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Analyze the formation of the image by a bi-convex lens with a radius of curvature of 20 cm when an object of height 2 cm is placed 30 cm away. Discuss the characteristics of the image produced.
12. An observer moves past a meter stick at a velocity of half the speed of light. Discuss how relativistic effects influence the measured length of the meter stick and calculate the observed length.

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